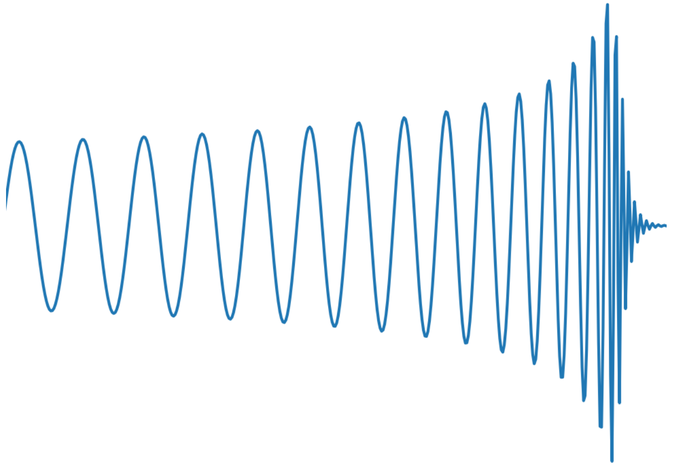
引力波数据分析



中国科学院紫金山天文台

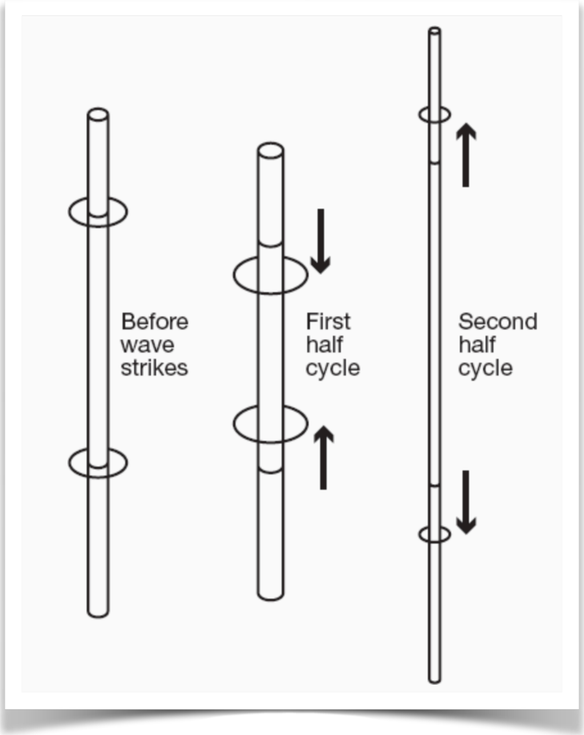
王一帆

2026.09.29

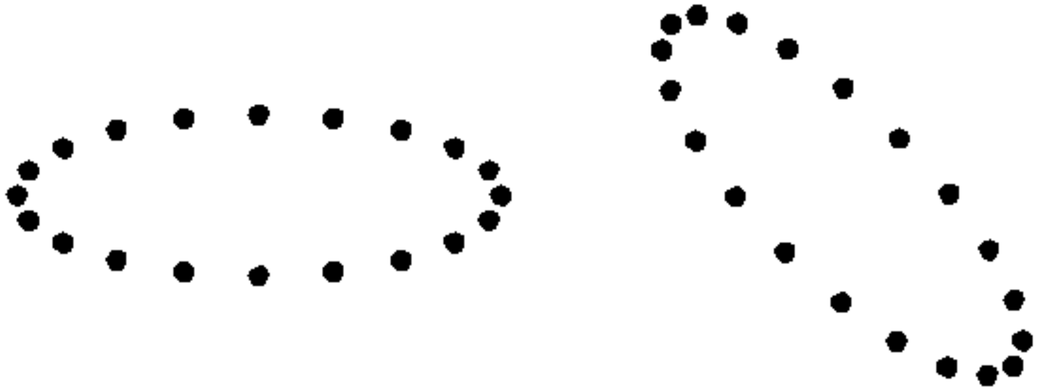
@引力波数据分析workshop

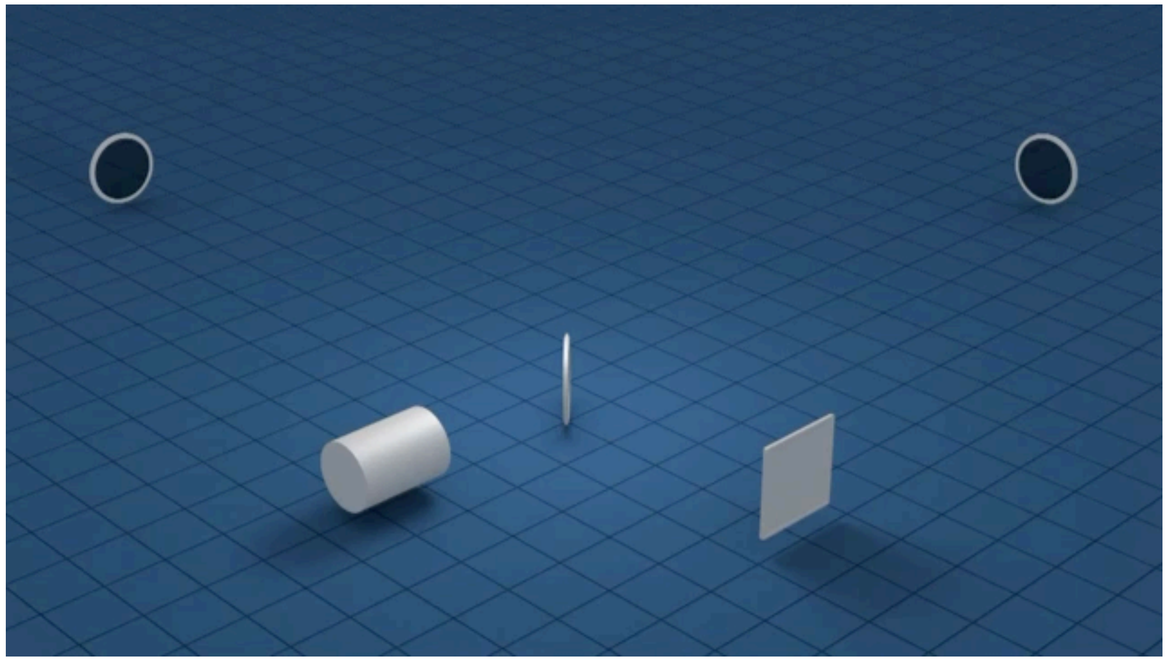
MAX-PLANCK-GESELLSCHAFT

- 1915: Albert Einstein founded General Relativity
- 1916: Einstein predicted GW; 1936: Einstein: GW doesn't exist
- 1957: Bondi thought experiment
- 1968: The first attempts to detect gravitational waves using large-scale detectors such as the Weber Bar and the resonant-mass detector were made.



- 1970s: Rainer Weiss (et al.) pioneered the concept of using laser interferometers to measure gravitational waves





- 1990s: Laser Interferometer Gravitational-Wave Observatory (LIGO) in the US and Virgo in Europe, were funded and built.
  - Lee Finn "Detection, measurement, and gravitational radiation" (1992, https://arxiv.org/abs/gr-qc/9209010);
  - Template bank construction(1992, “Choice of filters for the detection of gravitational waves from coalescing binaries”, DOI: 10.1103/physrevd.44.3819);
  - Detection statistic for stochastic background (1996, https://arxiv.org/abs/gr-qc/9604033);
  - Effective One Body (1999, https://arxiv.org/abs/gr-qc/9811091); Numerical Relativity breakthrough in 2005 (https://arxiv.org/abs/gr-qc/0407110)
- 2002: The first LIGO science run took place, but no gravitational wave signals were detected.
  - chi-square discriminator for GW detection (2004, https://arxiv.org/abs/gr-qc/0405045);
  - Pipeline constructions for Initial LIGO science run (“LALSuite”, https://git.ligo.org/lscsoft/lalsuite)
- 2010: Advanced LIGO and Advanced Virgo were proposed and designed to improve the sensitivity of gravitational wave detectors.
  - "find chirp" algorithm (2012, https://arxiv.org/abs/gr-qc/0509116), PyCBC (2015, https://arxiv.org/abs/1508.02357);

First direct observation of GWs from a BBH system on September 14, 2015:

GW150914

PRL **116,** 061102 (2016)

Selected for a Viewpoint in *Physics*
PHYSICAL REVIEW LETTERS

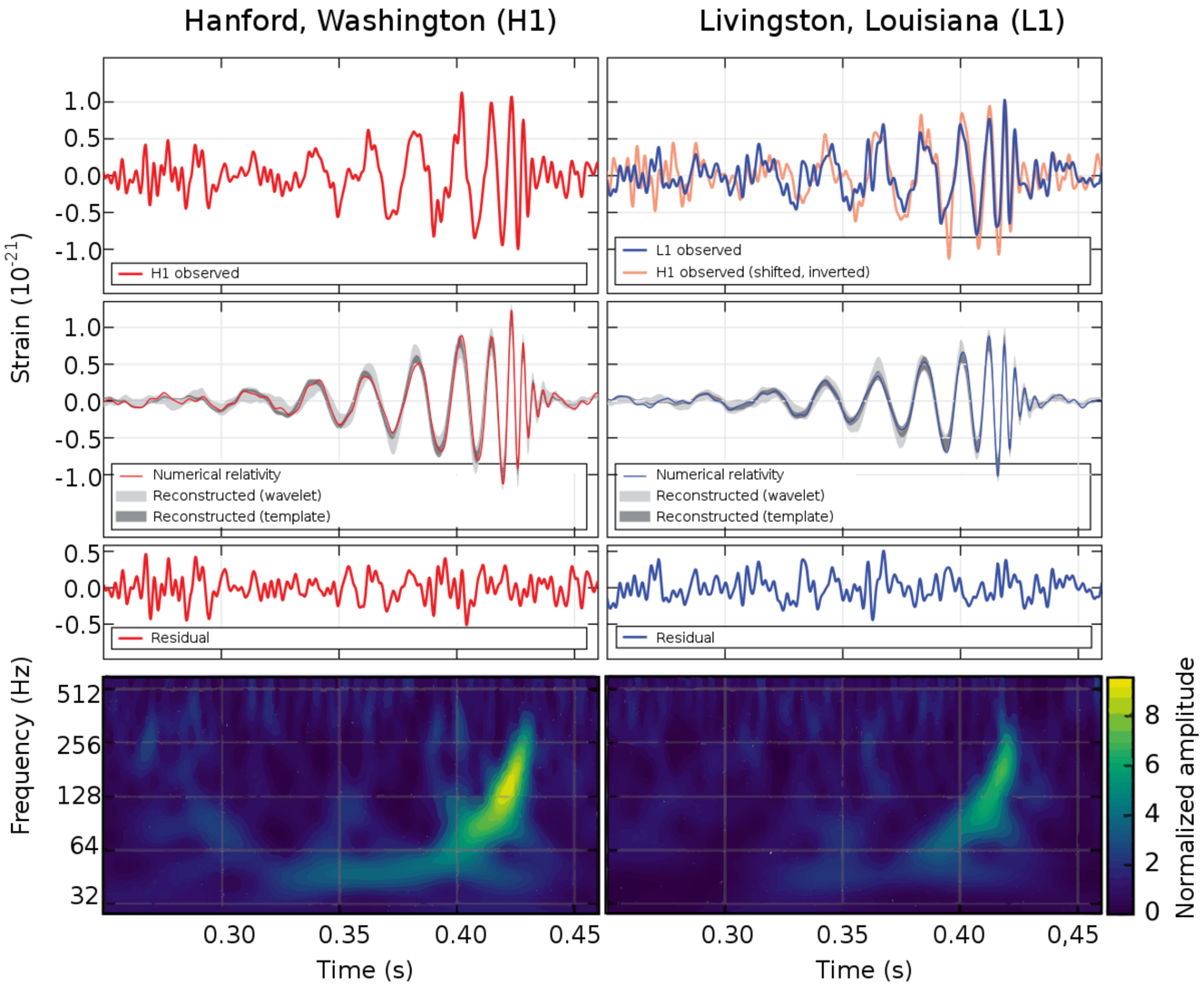
week ending
12 FEBRUARY 2016

Observation of Gravitational Waves from a Binary Black Hole Merger

B. P. Abbott *et al.*[*]
(LIGO Scientific Collaboration and Virgo Collaboration)
(Received 21 January 2016; published 11 February 2016)

On September 14, 2015 at 09:50:45 UTC the two detectors of the Laser Interferometer Gravitational-Wave Observatory simultaneously observed a transient gravitational-wave signal. The signal sweeps upwards in frequency from 35 to 250 Hz with a peak gravitational-wave strain of 1.0×10^{-21}. It matches the waveform predicted by general relativity for the inspiral and merger of a pair of black holes and the ringdown of the resulting single black hole. The signal was observed with a matched-filter signal-to-noise ratio of 24 and a false alarm rate estimated to be less than 1 event per 203 000 years, equivalent to a significance greater than 5.1σ. The source lies at a luminosity distance of 410^{+160}_{-180} Mpc corresponding to a redshift $z = 0.09^{+0.03}_{-0.04}$. In the source frame, the initial black hole masses are $36^{+5}_{-4}M_{\odot}$ and $29^{+4}_{-4}M_{\odot}$, and the final black hole mass is $62^{+4}_{-4}M_{\odot}$, with $3.0^{+0.5}_{-0.5}M_{\odot}c^2$ radiated in gravitational waves. All uncertainties define 90% credible intervals.

Phys. Rev. Lett. **116**, 061102 (2016)

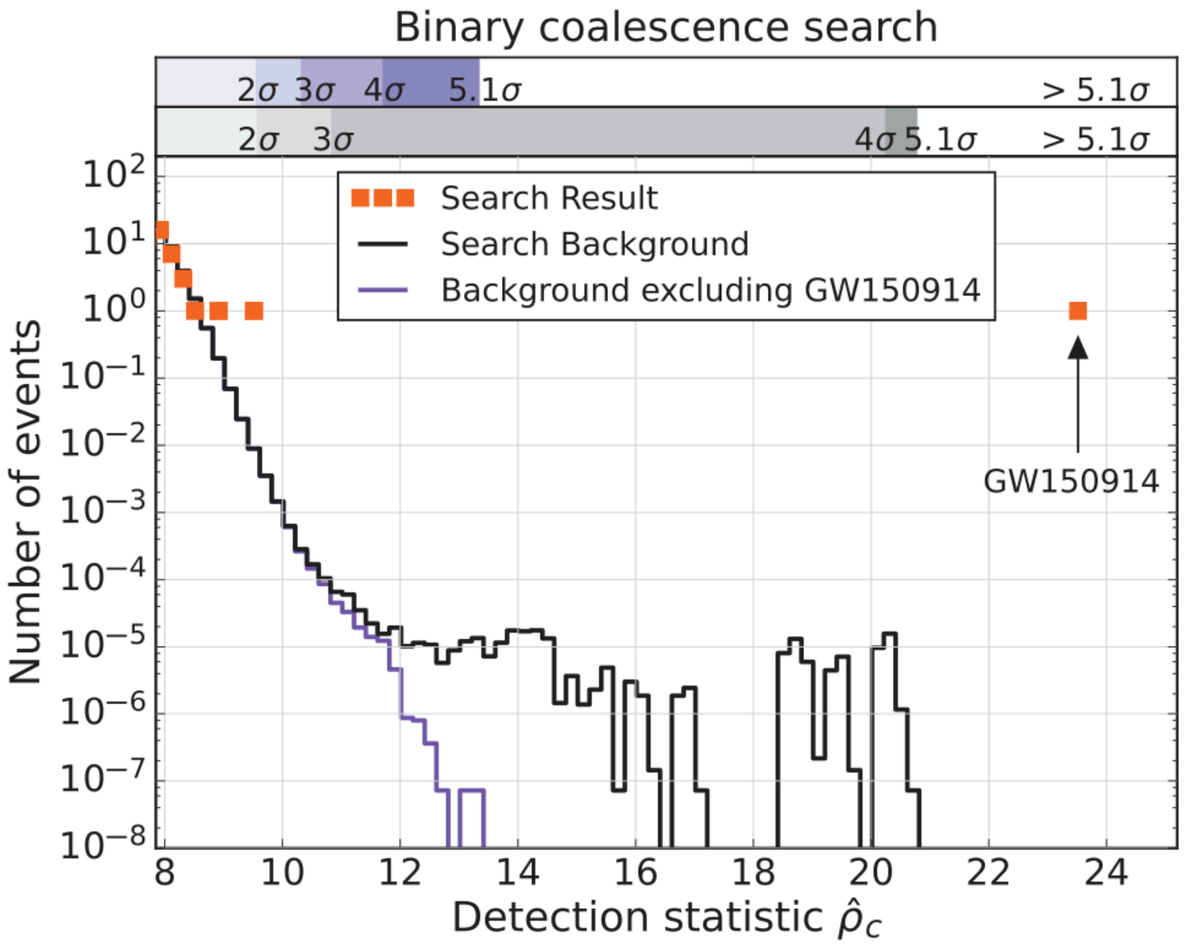


First direct observation of GWs from a BBH system on September 14, 2015:

GW150914

TABLE I. Source parameters for GW150914. We report median values with 90% credible intervals that include statistical errors, and systematic errors from averaging the results of different waveform models. Masses are given in the source frame; to convert to the detector frame multiply by $(1+z)$ [90]. The source redshift assumes standard cosmology [91].

Primary black hole mass	$36^{+5}_{-4} M_\odot$
Secondary black hole mass	$29^{+4}_{-4} M_\odot$
Final black hole mass	$62^{+4}_{-4} M_\odot$
Final black hole spin	$0.67^{+0.05}_{-0.07}$
Luminosity distance	410^{+160}_{-180} Mpc
Source redshift z	$0.09^{+0.03}_{-0.04}$



First direct observation of GWs from a BI system on August 17, 2017:

GW170817

PRL **119,** 161101 (2017) Selected for a Viewpoint in *Physics* PHYSICAL REVIEW LETTERS week ending 20 OCTOBER 2017

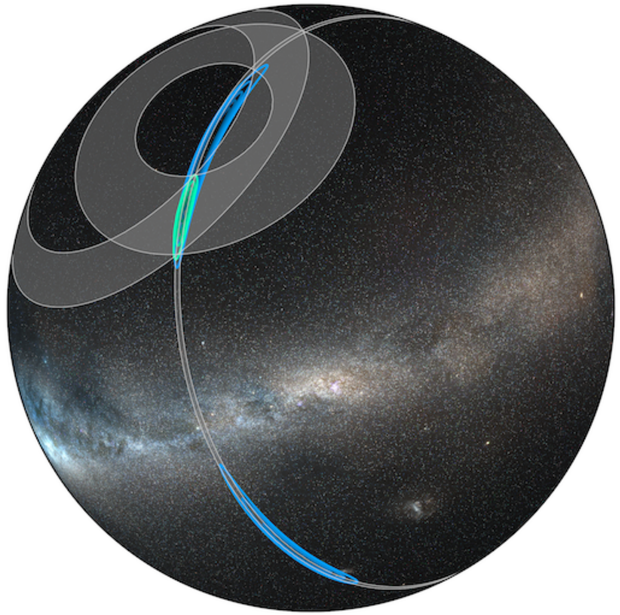
GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral

B. P. Abbott *et al.*[*]
(LIGO Scientific Collaboration and Virgo Collaboration)

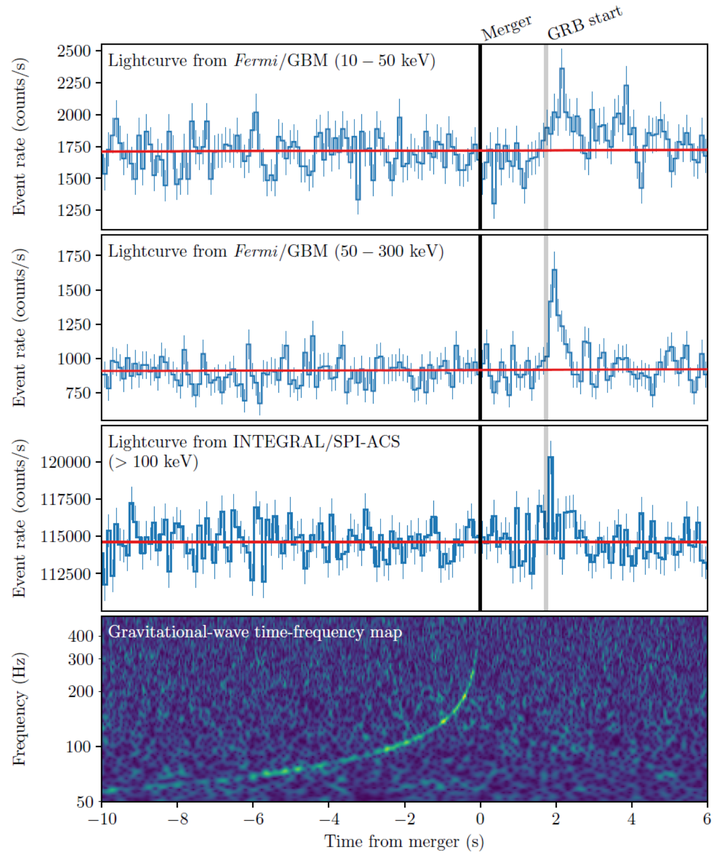
(Received 26 September 2017; revised manuscript received 2 October 2017; published 16 October 2017)

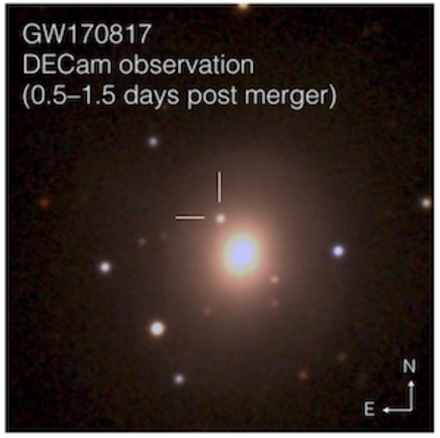
On August 17, 2017 at 12:41:04 UTC the Advanced LIGO and Advanced Virgo gravitational-wave detectors made their first observation of a binary neutron star inspiral. The signal, GW170817, was detected with a combined signal-to-noise ratio of 32.4 and a false-alarm-rate estimate of less than one per 8.0×10^4 years. We infer the component masses of the binary to be between 0.86 and 2.26 $M_\odot$, in agreement with masses of known neutron stars. Restricting the component spins to the range inferred in binary neutron stars, we find the component masses to be in the range 1.17–1.60 $M_\odot$, with the total mass of the system $2.74^{+0.04}_{-0.01} M_\odot$. The source was localized within a sky region of 28 deg^2 (90% probability) and had a luminosity distance of 40^{+8}_{-14} Mpc, the closest and most precisely localized gravitational-wave signal yet. The association with the γ-ray burst GRB 170817A, detected by Fermi-GBM 1.7 s after the coalescence, corroborates the hypothesis of a neutron star merger and provides the first direct evidence of a link between these mergers and short γ-ray bursts. Subsequent identification of transient counterparts across the electromagnetic spectrum in the same location further supports the interpretation of this event as a neutron star merger. This unprecedented joint gravitational and electromagnetic observation provides insight into astrophysics, dense matter, gravitation, and cosmology.

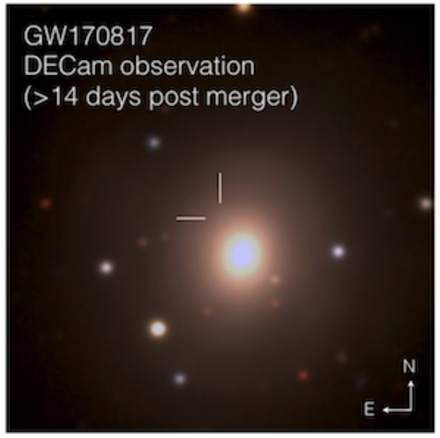
DOI: 10.1103/PhysRevLett.119.161101



Credit: LIGO/Virgo/NASA/Leo Singer (Milky Way image: Axel Mellinger)

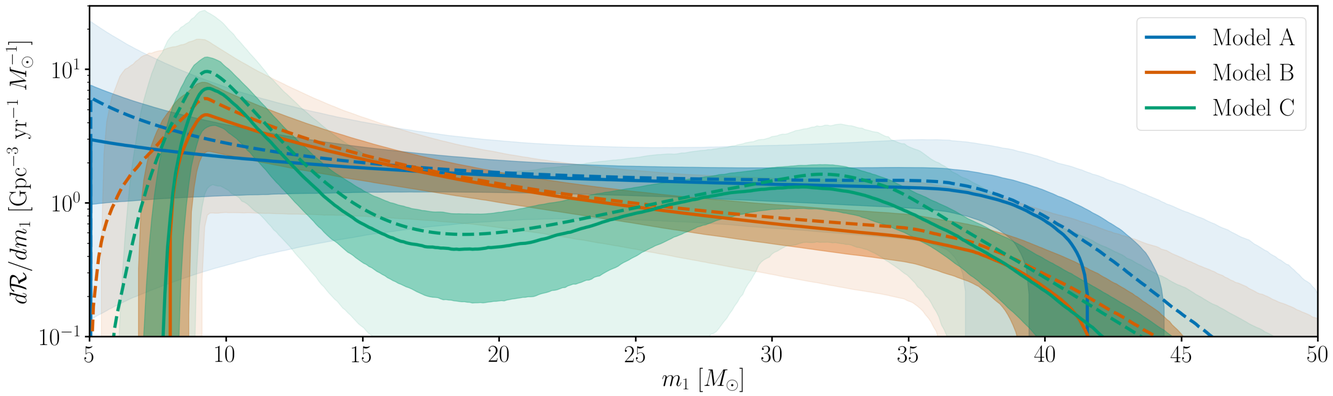




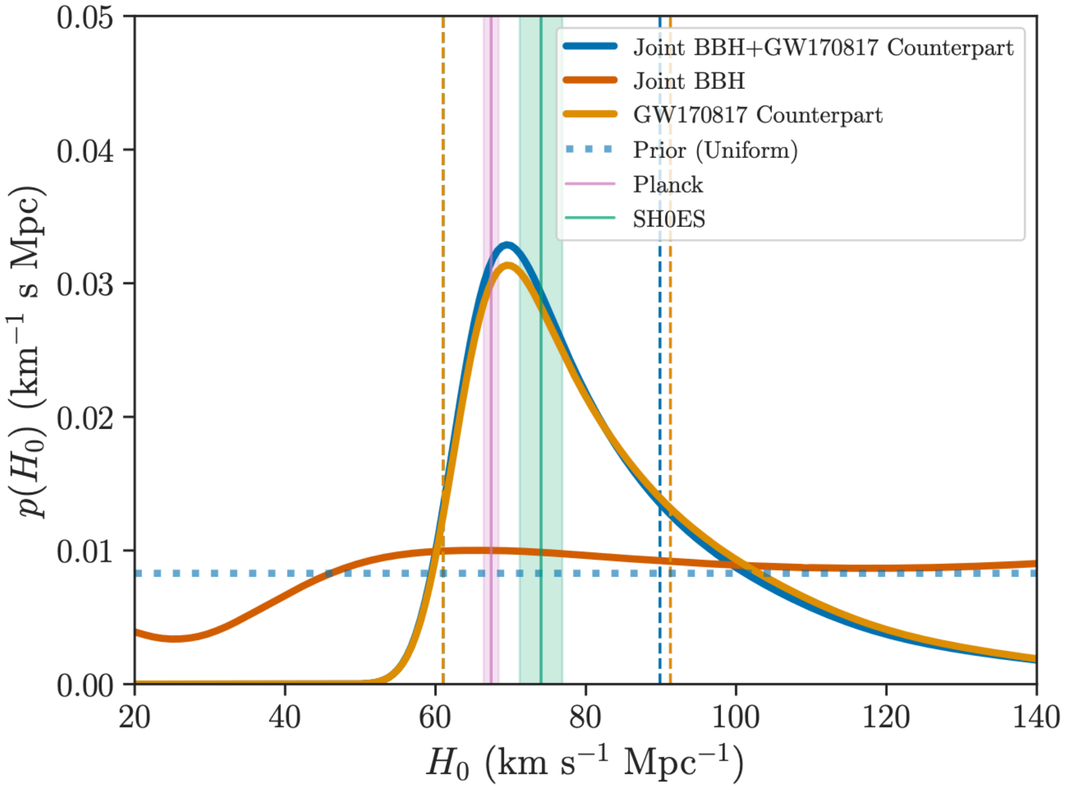


GW detections and analysis enable lots of interesting science:

- **astrophysical population models,**
- **cosmology,**
- **tests of General Relativity,**
- **matter in extreme states,**
- **and a lot more!**

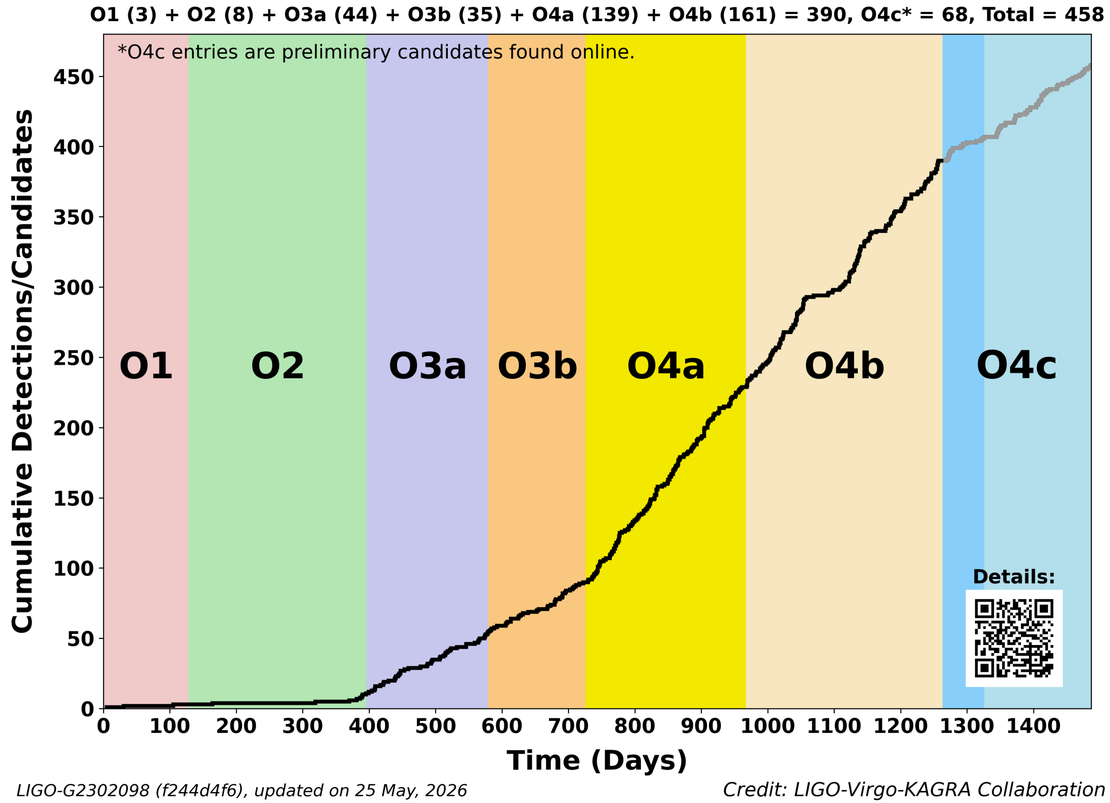


Astrophysical population of BHs. (Figure: rate of black holes of a given mass which participate in mergers over a given volume of space, from GWTC, LVK+2019)



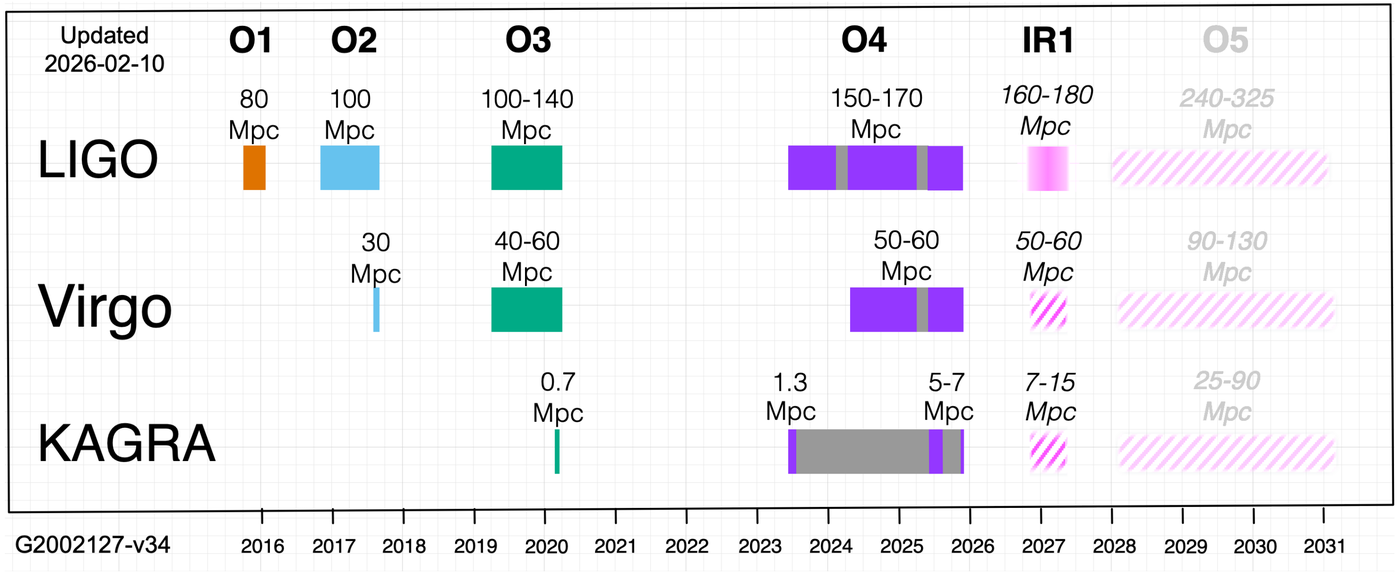
Hubble constant inference. (Figure: inference of the Hubble constant using GWs + BNS electromagnetic counterpart, from Abbott+2020)

Gravitational wave events & Observing Plan



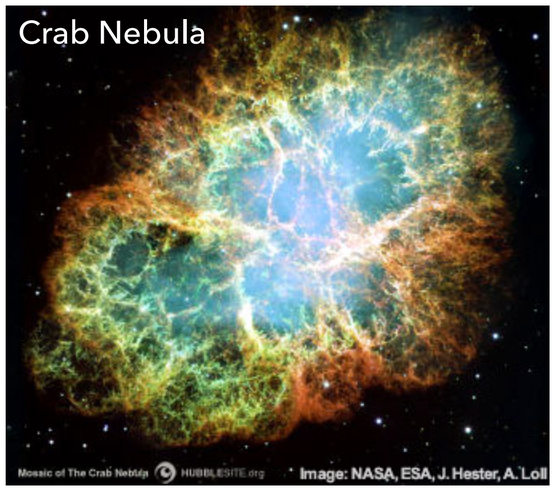
LIGO-G2302098 (f244d4f6), updated on 25 May, 2026

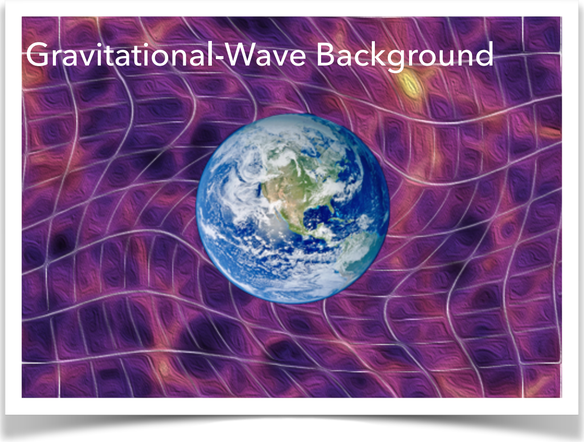
Credit: LIGO-Virgo-KAGRA Collaboration



Possible new type

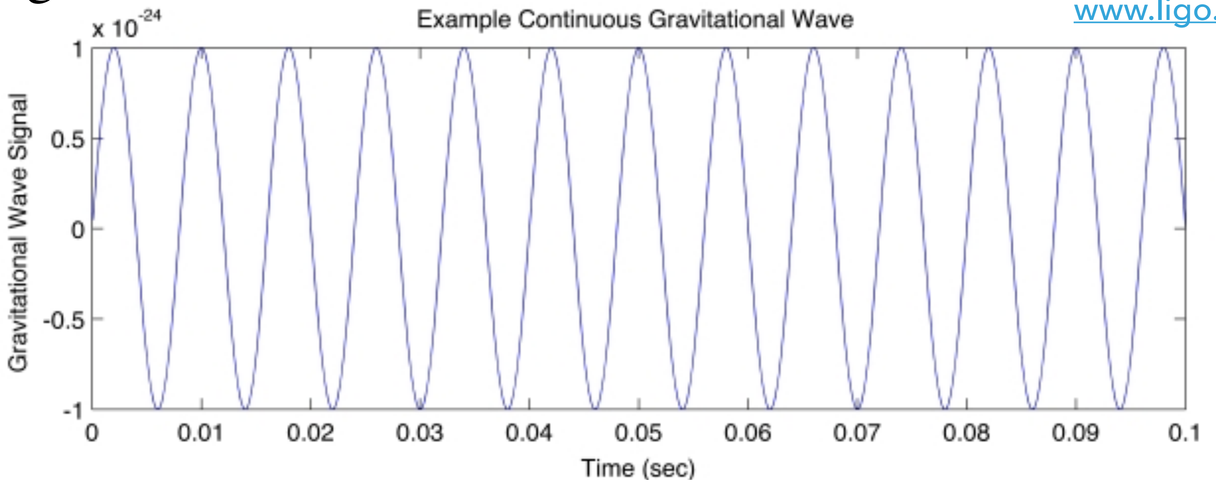
- More types of gravitational waves are yet to be discovered:
  - Burst (e.g. Supernova explosion)
  - Continuous Wave (e.g. Spinning neutron star)
  - Stochastic Gravitational-Wave Background

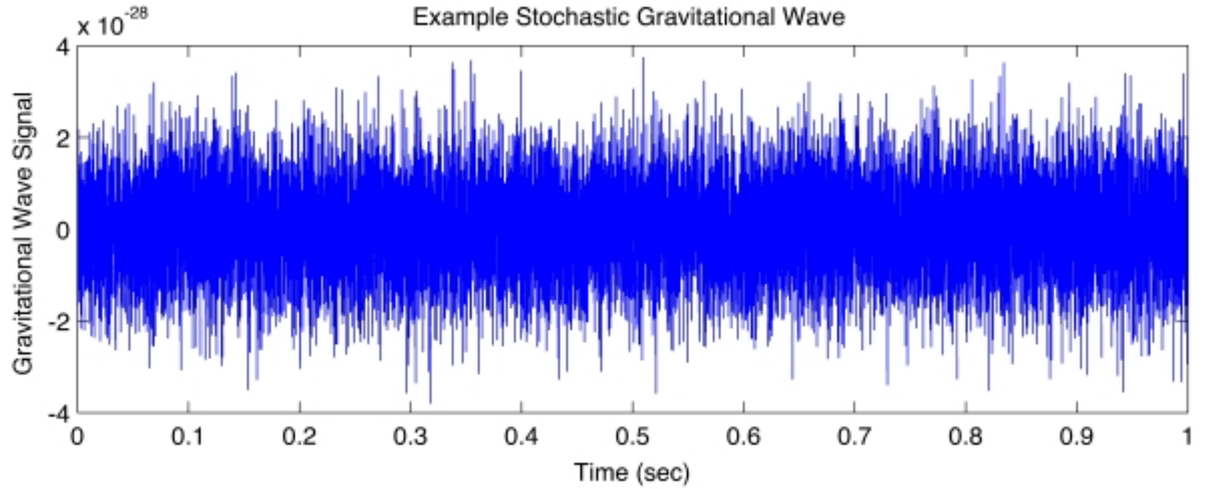


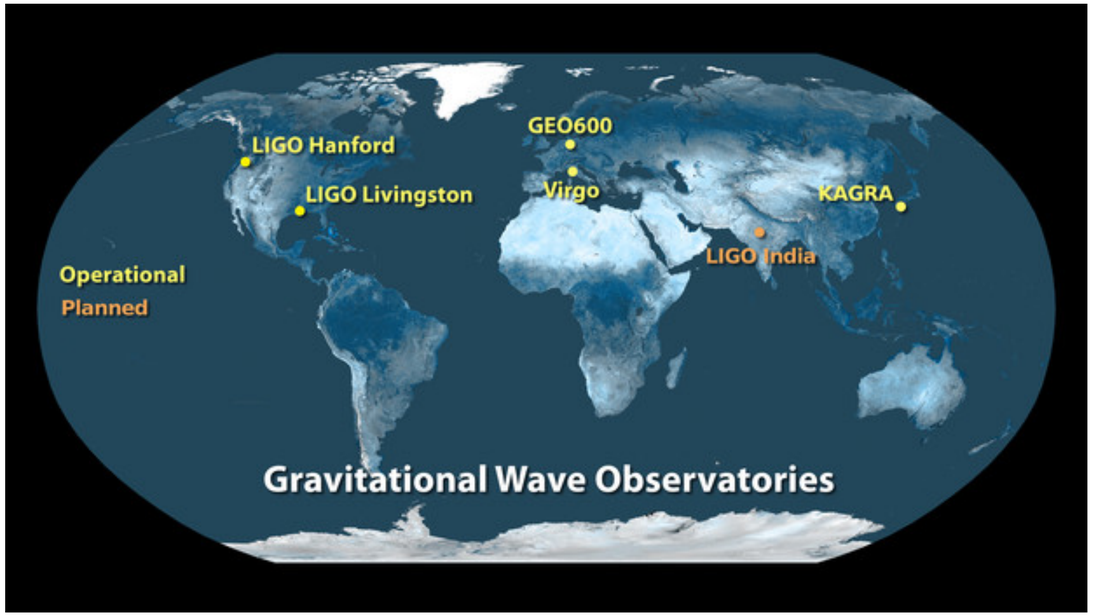


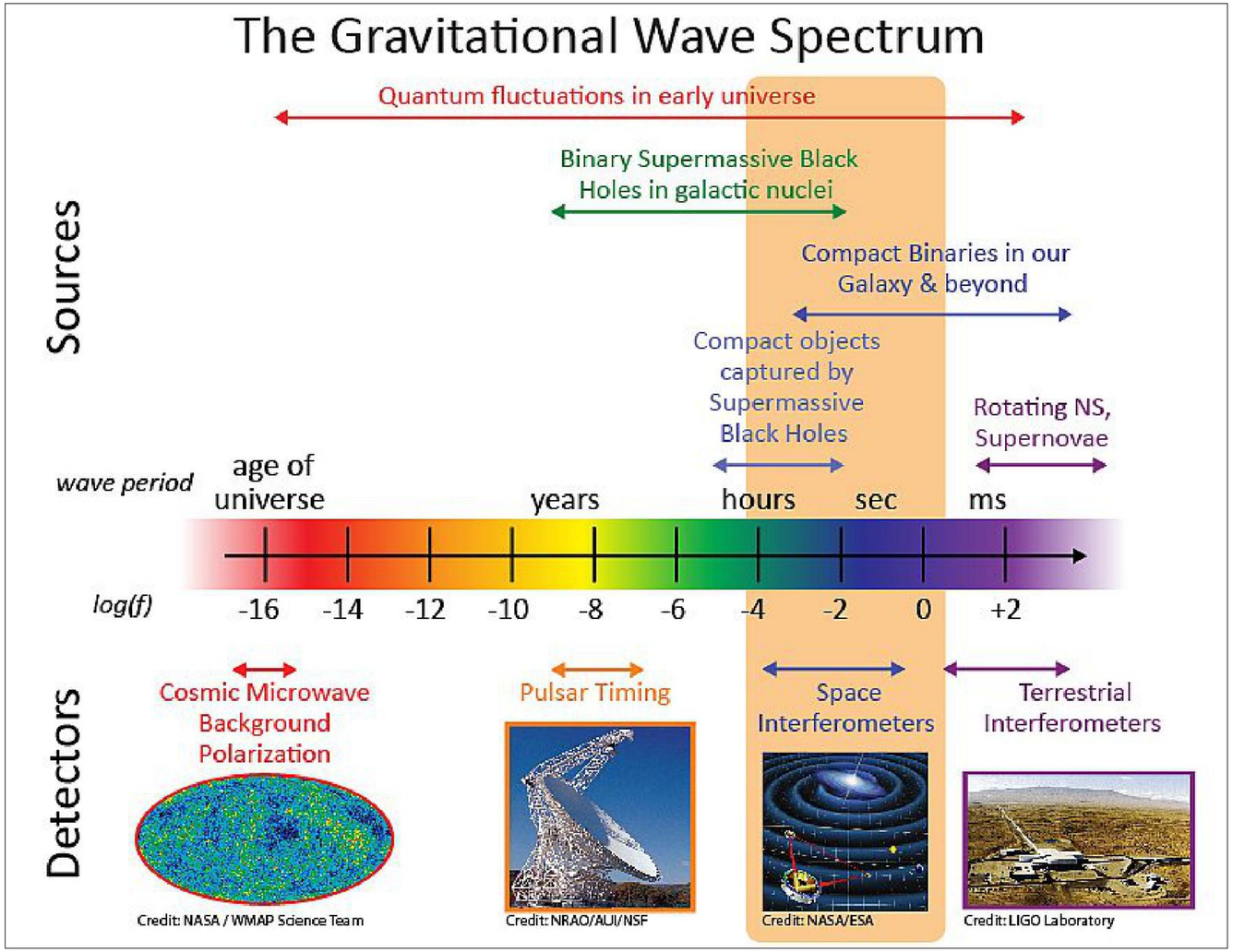
made by Yifan

www.ligo.org/science

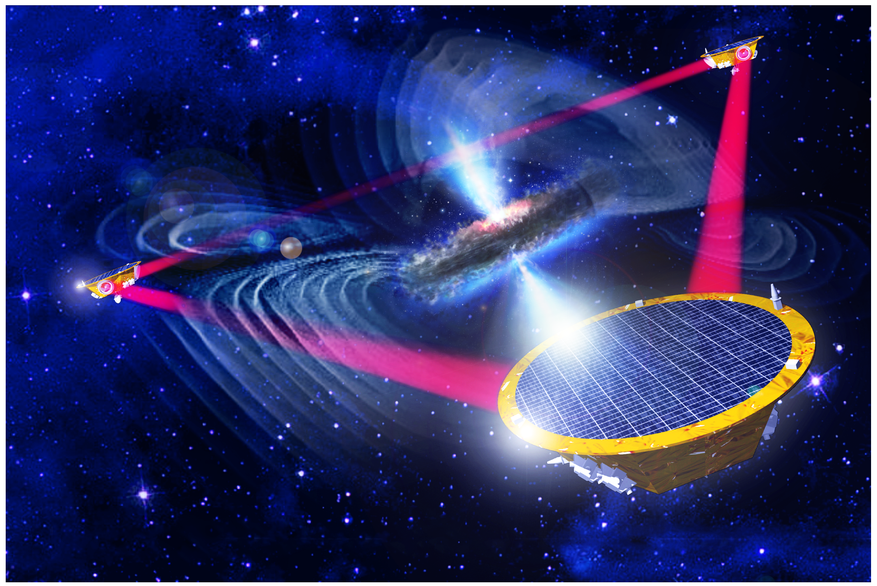








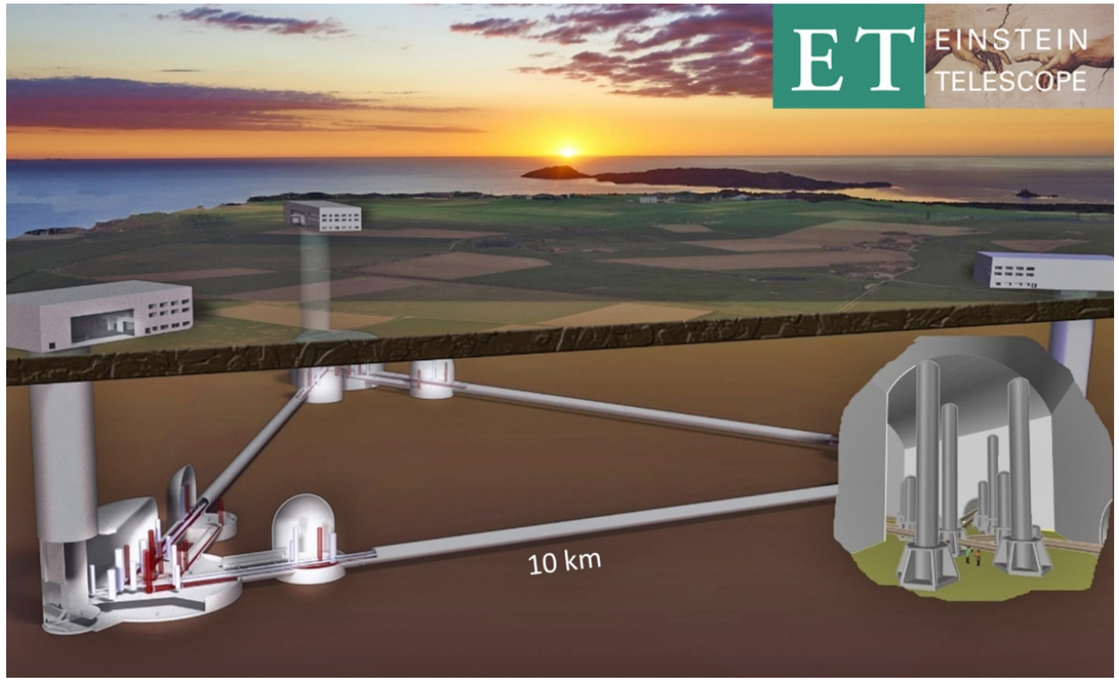
其他探测引力波的方式

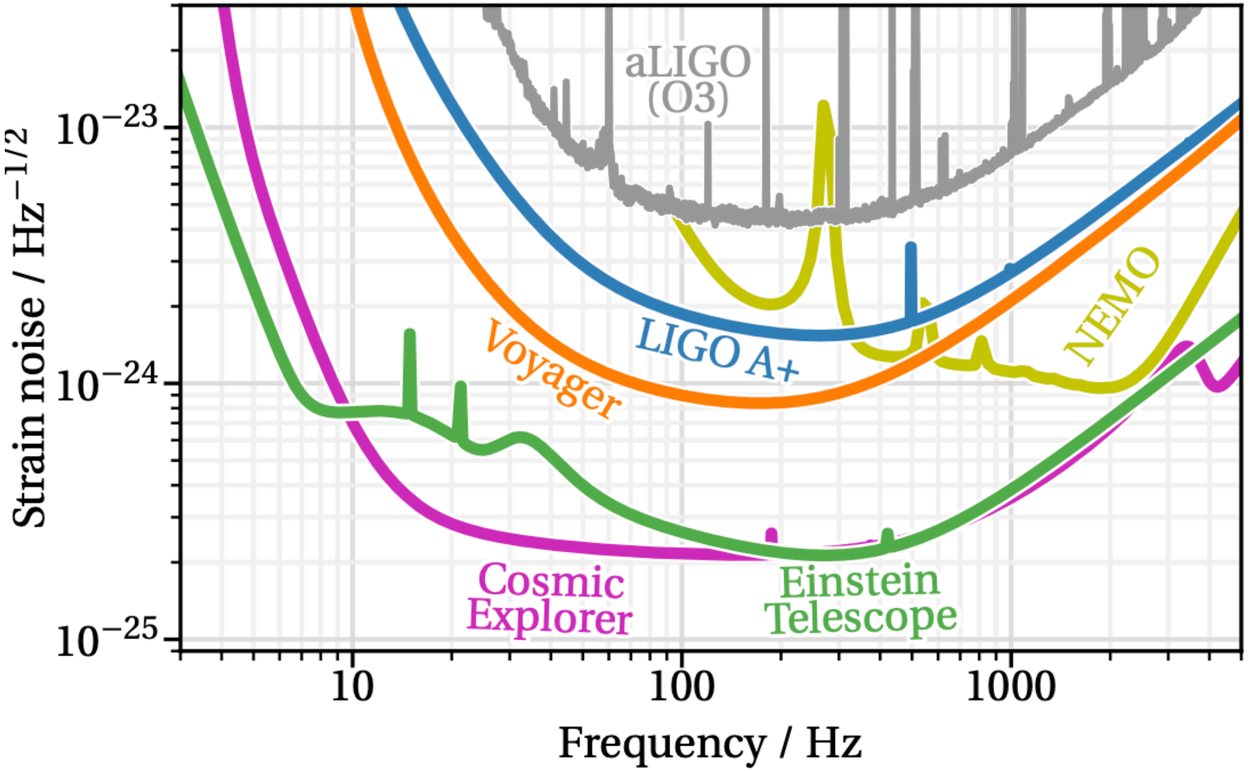


空间引力波探测器（~2035）

Future detection plan

- Europe: Einstein Telescope
- US: Cosmic Explorer
- Australia: NEMO (Neutron Star Extreme Matter Observatory)

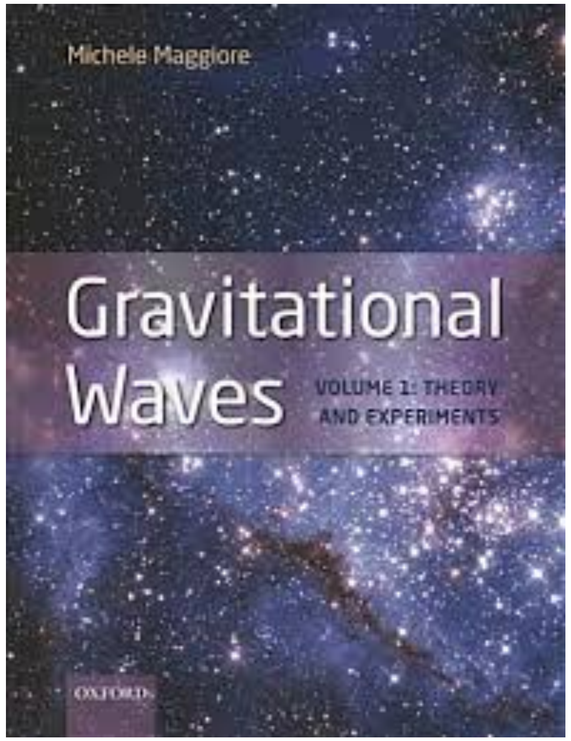


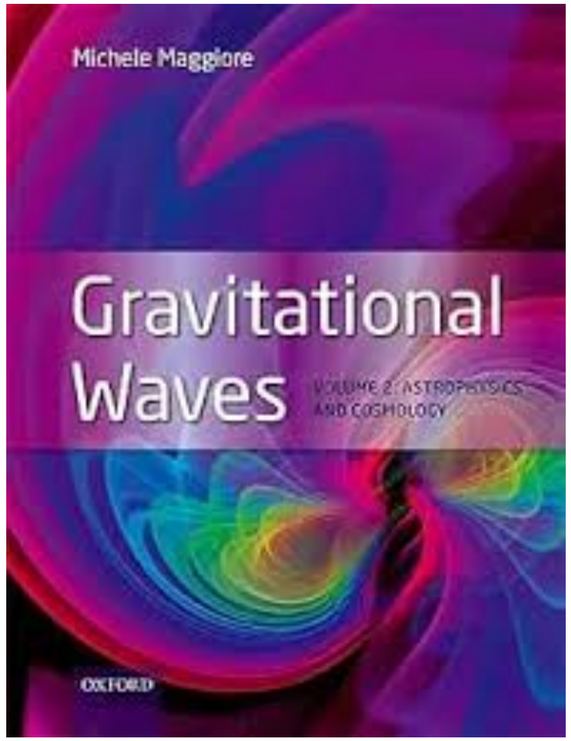


Complementary Reading:

Michele Maggiore ``Gravitational Waves: Volume 1: Theory and Experiments`` Sec 5: Intro to data analysis

``Gravitational Waves. Vol. 2: Astrophysics and Cosmology``





PhD Thesis

Lecture Notes by Yifan: https://github.com/yi-fan-wang/JESS2026